

COST SENSITIVE LEARNING FOR MYOCARDIAL INFARCTION MORTALITY PREDICTION CRISP-DM PROCESS REPORTs

BUSINESS UNDERSTANDING

Problems of real-life complexity are needed to test and compare various data mining and pattern recognition methods. The proposed database can be used to solve two practically important problems: predicting complications of Myocardial Infarction (MI) based on information about the patient (i) at the time of admission and (ii) on the third day of the hospital period. Another important group of tasks is phenotyping of disease (cluster analysis), dynamic phenotyping (filament extraction and identification of disease trajectories) and visualisation (disease mapping).

MI is one of the most challenging problems of modern medicine. Acute myocardial infarction is associated with high mortality in the first year after it. The incidence of MI remains high in all countries. This is especially true for the urban population of highly developed countries, which is exposed to chronic stress factors, irregular and not always balanced nutrition. In the United States, for example, more than a million people suffer from MI every year, and 200-300 thousand of them die from acute MI before arriving at the hospital.

The course of the disease in patients with MI is different. MI can occur without complications or with complications that do not worsen the long-term prognosis. At the same time, about half of patients in the acute and subacute periods have complications that lead to worsening of the disease and even death. Even an experienced specialist can not always foresee the development of these complications. In this regard, predicting complications of myocardial infarction in order to timely carry out the necessary preventive measures is an important task.

DATA UNDERSTANDING: *Data Description Report*

- **Location:** The original dataset is available at <https://archive-beta.ics.uci.edu/dataset/579/myocardial+infarction+complications>
- **Method used to acquire:** Download the dataset using the `ucimlrepo` library, which provides a simple way to interact with uci machine learning resources.
- **Format:** csv file
- **Quantity:** Rows (1700) - Columns (124)
- **Identities of the fields:** The dataset analyzed here was collected in the Krasnoyarsk Interdistrict Clinical Hospital (Russia) in 1992-1995 years, but has only recently been deposited to the public domain. It contains information about 1700 patients and 110 features characterizing the clinical phenotypes and 12 features representing possible complications of the myocardial infarction disease. In general columns 2-112 can be used as input data for prediction. Possible complications (outputs) are listed in columns 113-124. There are four possible time moments for complication prediction on base of the information known at:
 - the time of admission to hospital - all input columns (2-112) except 93, 94, 95, 100, 101, 102, 103, 104, 105 can be used for prediction;
 - the end of the first day (24 hours after admission to the hospital): all input columns (2-112) except 94, 95, 101, 102, 104, 105 can be used for prediction;
 - the end of the second day (48 hours after admission to the hospital) all input columns (2-112) except 95, 102, 105 can be used for prediction;
 - the end of the third day (72 hours after admission to the hospital) all input columns (2-112) can be used for prediction.
- **Surfaces features:** ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, IBS_NASL, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE, TIME_B_S, R_AB_1_n, R_AB_2_n, R_AB_3_n, NA_KB, NOT_NA_KB, LID_KB, NITR_S, NA_R_1_n, NA_R_2_n, NA_R_3_n, NOT_NA_1_n, NOT_NA_2_n, NOT_NA_3_n, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n,
 - **Quantitative Variables**
 - **Discrete:** AGE, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT
 - **Continuous:** K_BLOOD, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE
 - **Categorical Variables**
 - **Nominal**
 - **Binary:** SEX, IBS_NASL, SIM_GIPERT, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, GIPER_NA, NA_KB, NOT_NA_KB, LID_KB, NITR_S, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n,

- TIKL_S_n, TRENT_S_n, FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN
 - **General:** LET_IS, ID, IBS_POST
 - **Ordinal:** INF_ANAM, STENOK_AN, FK_STENOK, GB, DLIT_AG, ZSN_A, TIME_B_S, R_AB_1_n, R_AB_2_n, R_AB_3_n, NA_R_1_n, NA_R_2_n, NA_R_3_n, NOT_NA_1_n, NOT_NA_2_n, NOT_NA_3_n, ant_im, lat_im, inf_im, post_im
- **Targets:** FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS
 - **Quantitative Variables**
 - **Discrete:** -
 - **Continuous:** -
 - **Categorical Variables**
 - **Nominal**
 - **Binary:** FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN
 - **General:** LET_IS
 - **Ordinal:** -
- **Features and Target by time moments for complication prediction:**
 - **admission**
 - **included:** ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, IBS_NASL, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE, TIME_B_S, NA_KB, NOT_NA_KB, LID_KB, NITR_S, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n, FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS
 - **excluded:** R_AB_1_n, R_AB_2_n, R_AB_3_n, NA_R_1_n, NA_R_2_n, NA_R_3_n, NOT_NA_1_n, NOT_NA_2_n, NOT_NA_3_n
 - **end of first day (after 24 hours)**
 - **included:** ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, IBS_NASL, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE, TIME_B_S, R_AB_1_n, NA_KB, NOT_NA_KB, LID_KB, NITR_S,

NA_R_1_n, NOT_NA_1_n, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n, FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS

- **excluded:** R_AB_2_n, R_AB_3_n, NA_R_2_n, NA_R_3_n, NOT_NA_2_n, NOT_NA_3_n

◦ **end of second day (after 48 hours)**

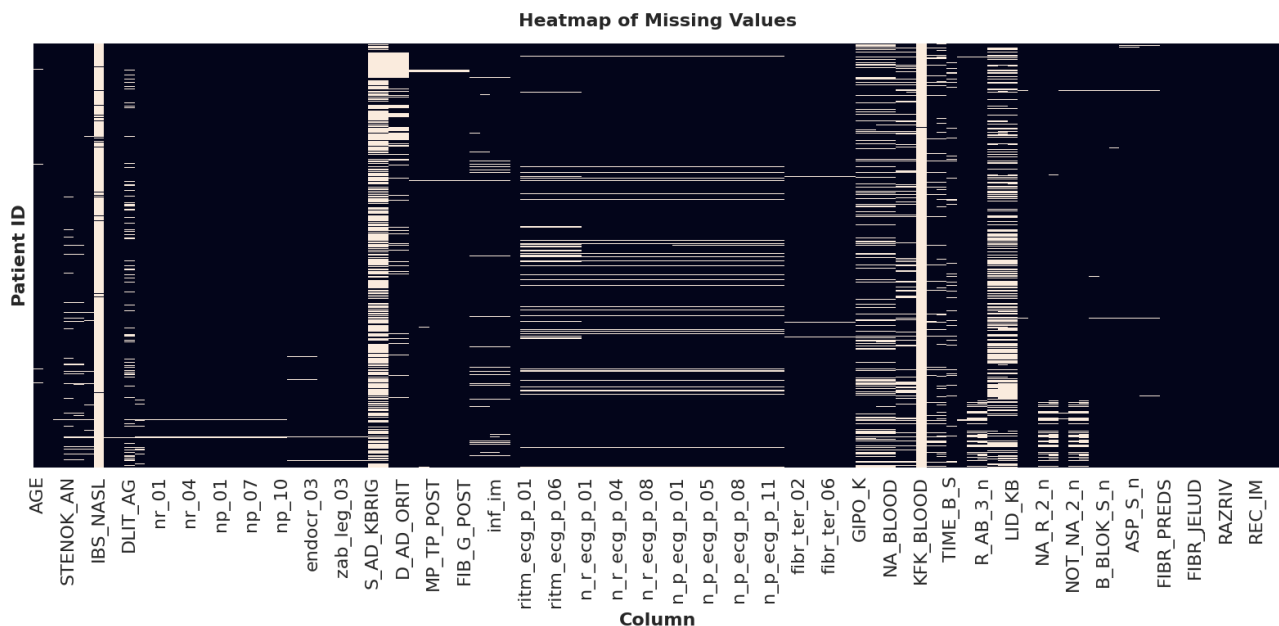
- **included:** ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, IBS_NASL, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE, TIME_B_S, R_AB_1_n, R_AB_2_n, NA_KB, NOT_NA_KB, LID_KB, NITR_S, NA_R_1_n, NA_R_2_n, NOT_NA_1_n, NOT_NA_2_n, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n, FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS
- **excluded:** R_AB_3_n, NA_R_3_n, NOT_NA_3_n

◦ **end of third day (after 72 hours)**

- **included:** ALL
 - **excluded:** -
-

DATA UNDERSTANDING: *Data Quality Report*

- **Missing values Analysis**



- **Row-Level Analysis:** Total rows=1700, rows with ≥ 1 missing value=1700/1700 (100.00/100.00%). First of all, we quantify the general missingness in the dataset: the best-case record completeness has 0.81% of missing values; the worst-case record has 48.78% of missing values; the average quantity of missing values is 7.64%. Since 100% of rows have at least one missing value, the next step is to measure severity of missingness per row. We fix different threshold of missingness to assess how many rows have a level of missingness greater than the threshold: with a threshold $> 5\%$ we have 992/1700 record with a level of missingness greater than 5%; with a threshold $> 20\%$ we have 134/1700 record with a level of missingness greater than 5%; with a threshold $> 50\%$ we have 0/1700 record with a level of missingness greater than 50%.
- **Column-Level Analysis:** We check the percentage of missingness in each columns to identify columns with an high level of missingness (to delete) and columns with moderate/low level of missing values (imputation). We create different bins to identify how many columns have a level of missing values which falls in those bins:
 - (0-1] % $\rightarrow$ 37 columns
 - (1-5] % $\rightarrow$ 21 columns
 - (5-10] % $\rightarrow$ 35 columns
 - (10-20] % $\rightarrow$ 6 columns
 - (20-50] % $\rightarrow$ 7 columns
 - (50-80] % $\rightarrow$ 2 columns
 - (80-100] % $\rightarrow$ 2 columns

In the data cleaning phase we will address the problem of missing values to select what record will be deleted, what features will be deleted, what missing values could be handle by imputation, what type of imputation is the most proper one.

- **Duplicated Row:** Number of duplicate rows: 0/1700 (0.000%). Nothing to do.

- **Class Balance:** In general, the dataset presents different possible targets, each of these representing one possible complication, with an additional target representing the lethal outcome. We report the distribution of patients for each possible target and value:

```
=====
SAMPLES DISTRIBUTION BY: FIBR_PREDS
=====
- Class 0 - NO: 1530 (90.000%)
- Class 1 - YES: 170 (10.000%)
=====

SAMPLES DISTRIBUTION BY: PREDS_TAH
=====
- Class 0 - NO: 1680 (98.824%)
- Class 1 - YES: 20 (1.176%)
=====

SAMPLES DISTRIBUTION BY: JELUD_TAH
=====
- Class 0 - NO: 1658 (97.529%)
- Class 1 - YES: 42 (2.471%)
=====

SAMPLES DISTRIBUTION BY: A_V_BLOK
=====
- Class 0 - NO: 1643 (96.647%)
- Class 1 - YES: 57 (3.353%)
=====

SAMPLES DISTRIBUTION BY: OTEK_LANC
=====
- Class 0 - NO: 1541 (90.647%)
- Class 1 - YES: 159 (9.353%)
=====

SAMPLES DISTRIBUTION BY: RAZRIV
=====
- Class 0 - NO: 1646 (96.824%)
- Class 1 - YES: 54 (3.176%)
=====

SAMPLES DISTRIBUTION BY: DRESSLER
=====
- Class 0 - NO: 1625 (95.588%)
- Class 1 - YES: 75 (4.412%)
=====

SAMPLES DISTRIBUTION BY: ZSN
=====
- Class 0 - NO: 1306 (76.824%)
- Class 1 - YES: 394 (23.176%)
=====

SAMPLES DISTRIBUTION BY: REC_IM
=====
- Class 0 - NO: 1541 (90.647%)
- Class 1 - YES: 159 (9.353%)
=====

SAMPLES DISTRIBUTION BY: P_IM_STEN
=====
- Class 0 - NO: 1552 (91.294%)
- Class 1 - YES: 148 (8.706%)
=====
```

```

=====
SAMPLES DISTRIBUTION BY: LET_IS
=====
- Class 0 - ALIVE: 1429 (84.059%)
- Class 1 - CARDIOGENIC SHOCK: 110 ( 6.471%)
- Class 2 - PULMONARY EDEMA: 18 ( 1.059%)
- Class 3 - MYOCARDIAL RUPTURE: 54 ( 3.176%)
- Class 4 - CONGESTIVE HEART FAILURE: 23 ( 1.353%)
- Class 5 - THROMBOEMBOLISM: 12 ( 0.706%)
- Class 6 - ASYSTOLE: 27 ( 1.588%)
- Class 7 - VENTRICULAR FIBRILLATION: 27 ( 1.588%)
=====
SAMPLES DISTRIBUTION BY: LET_IS (BINARY)
=====
- Class 0 - ALIVE: 1429 (84.059%)
- Class 1 - DEATH: 271 (15.941%)
=====

```

As reported, the dataset suffers of severe imbalance with respect to every single possible targets. In particular, the LET_IS target, even if we choose to collapse in just two class (ALIVE / DEATH), still remains with a severe imbalance.

Whatever is the target we choose to use to perform the analysis, we should address the imbalance with different techniques (data-level, algorithm-level, cost-sensitive-learning). We will address this problem in the modelling phase.

DATA UNDERSTANDING: *Data Exploration Report*

- Class Balance:
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DATA PREPARATION: *Data Sampling Report*

DATA PREPARATION: *Data Cleaning Report*

DATA PREPARATION: *Feature Selection Report*

DATA PREPARATION: *Data Construction Report*

DATA PREPARATION: *Data Integration Report*
NOTHING TO REPORT

DATA PREPARATION: *Data Format Report*
NOTHING TO REPORT

MODELING
NOTHING TO REPORT

EVALUATION
NOTHING TO REPORT

DEPLOYMENT
NOTHING TO REPORT
